

Assignment 1

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1 MAX-CFSU

Question 1: A Collision-Free Sensor Upload [10 marks]

A network of remote environmental sensors shares a collection of wireless gateways. During a short emergency-upload window, the system must choose a subset of the sensors to transmit. Every gateway j can hear exactly $k \geq 2$ distinct sensors; let B_j denote the set of such sensors for gateway j . (Note the sets B_j may overlap, i.e., for any two gateways $j \neq j'$, $B_j \cap B_{j'}$ need not be empty.)

Gateway j successfully decodes a packet if **exactly one** sensor in B_j transmits. If none transmits, the gateway receives no packet; if two or more transmit, their packets collide. Successfully serving gateway j has importance $p_j \geq 0$.

The objective is to choose a set of transmitting sensors that maximizes the total importance of the gateways that successfully decode a packet. Let OPT denote the maximum possible importance.

Design a randomized polynomial-time algorithm whose expected value is at least OPT/ e . Your solution should determine and justify the transmission probability used by the algorithm, prove the approximation guarantee, and state the running time.

Answer

Let I be the sensor set and J the gateway set, For each $j \in J$, let $B_j \subseteq I$ be the sensors it can hear, where $|B_j| = k \geq 2$, and $p_j \geq 0$ be the importance of gateway j .

Input: Sensor set I , gateway sets $(B_j)_{j \in J}$, importance weights $(p_j)_{j \in J}$, probability q

Output: Transmitting sensor set $X \subseteq I$

$X \leftarrow \emptyset$;

foreach $s \in I$ **do**

 Independently set $Z_s \leftarrow 1$ at probability q ;

if $Z_s = 1$ **then**

$X \leftarrow X \cup \{s\}$;

return X ;

1.1 Approximation factor

Proof. Define indicator binary variable τ as

$$\tau(j) = \begin{cases} 1, & j \text{ successfully decode} \\ 0, & \text{otherwise} \end{cases}$$

For each gateway, the probability of successful transmission can be expressed as

$$P(j) = q \cdot (1 - q)^{k-1} \cdot k$$

Expectation of the importance from gateway j as

$$\mathbb{E}[p_j \tau(j)] = p_j \cdot kq(1 - q)^{k-1}$$

and the total importance of all successfully transmitted gateway as

$$\mathbb{E}[\text{importance}(\tau)] = \mathbb{E}\left[\sum_j \text{importance}(\tau(j))\right] = \sum_j p_j \cdot kq(1 - q)^{k-1}$$

Define $f(q) = kq(1 - q)^{k-1}$ for $q \in [0, 1]$. Since

$$\begin{aligned} f'(q) &= k[(1 - q)^{k-1} - q(k - 1)(1 - q)^{k-2}] \\ &= k(1 - kq)(1 - q)^{k-2} \end{aligned}$$

$f'(q) > 0$ for $0 < q < 1/k$, and $f'(q) < 0$ for $1/k < q < 1$, and $f(0) = f(1) = 0$, thus the maximum occurs at $q = 1/k$. substitute q with $\frac{1}{k}$,

$$f(q)_{\max} = \left(1 - \frac{1}{k}\right)^{k-1}$$

$$\left(1 - \frac{1}{k}\right)^{k-1} = \left(\frac{1}{1 + \frac{1}{k-1}}\right)^{k-1} \geq \frac{1}{e}$$

We have

$$\mathbb{E}[\text{importance}(\tau)] \geq \frac{1}{e} \sum_j p_j \geq \frac{1}{e} OPT$$

□

1.2 Algorithm

Thus the algorithm is

Input: Sensor set I , gateway sets $(B_j)_{j \in J}$, importance weights $(p_j)_{j \in J}$
Output: Transmitting sensor set $X \subseteq I$
 $X \leftarrow \emptyset$;
foreach $s \in I$ **do**
 Independently set $Z_s \leftarrow 1$ at probability $\frac{1}{k}$;
 if $Z_s = 1$ **then**
 $X \leftarrow X \cup \{s\}$;
return X ;

1.3 Running time

Let

$$n = |I|, \quad g = |J|$$

The algorithm makes n independent random choices and takes $O(n)$ time, including reading the gateway sets gives $O(n + gk)$

2 PBS

Question 2: Planning a Biodiversity Survey [10 marks]

A conservation agency is choosing field-survey routes. Let U be the set of species that might be observed, and let $a_u \geq 0$ be the conservation importance of species $u \in U$. Suppose there are m routes. Route i costs c_i , observes a subset $S_i \subseteq U$, and may overlap with other routes. The agency has budget B , and every route satisfies $0 < c_i \leq B$. A species contributes its importance at most once, regardless of how many chosen routes observe it.

For a set A of routes, define its **value** as follows:

$$F(A) = \sum_{u \in \bigcup_{i \in A} S_i} a_u.$$

The goal is to maximize $F(A)$ subject to $\sum_{i \in A} c_i \leq B$.

Design a deterministic polynomial-time algorithm that returns a feasible subset of routes with value at least $\frac{1}{2} \left(1 - \frac{1}{e}\right) OPT$, where OPT is the value of an optimal budget-feasible subset. Prove the approximation guarantee and state the running time of your algorithm.

Answer

Define marginal value of adding a new route i to the collection of routes A

$$\Delta(i \mid A) = F(A \cup \{i\}) - F(A)$$

and the value density per cost of route i when being added to A as

$$\delta(i \mid A) = \frac{\Delta(i \mid A)}{c_i}$$

2.1 Algorithm

Input: Species set $U = \{u_1, u_2, \dots\}$, route set $\mathcal{S}$, budget B

Output: Route collection A

Stack $A \leftarrow \emptyset$;

while $B \geq 0 \wedge \mathcal{S} \neq \emptyset$ **do**

$i_{\max} = \arg \max_{i \in \mathcal{S}} \frac{\Delta(i|A)}{c_i}$;

$A.\text{push}(i_{\max})$;

$B \leftarrow B - c_{i_{\max}}$;

$\mathcal{S} \leftarrow \mathcal{S} \setminus \{i_{\max}\}$;

if $B < 0$ *# this guarantees feasibility* **then**

$i' \leftarrow A.\text{top}()$;

$A.\text{pop}()$;

return $\arg \max_{x \in \{A, \{i'\}\}} (F(x))$;

return A ;

2.2 Approximation factor proof

Define the following:

- $\mathcal{S}$: the set of routes, $|\mathcal{S}| = m$
- $S_i \subseteq U$: species contained in route i
- $A \subseteq \mathcal{S}$: chosen collection of routes
- $T(A) = \bigcup_{i \in A} S_i$: the set of species observed in A
- $F(A) = \sum_{u \in \bigcup_{i \in A} S_i} a_u$: total importance of A
- $c(A) = \sum_{i \in A} c_i$: total cost of A
- A^* : one optimal collection

As all importance is non negative, adding new routes never decreases total importance, meaning F is

$$OPT = F(A^*) \leq F(A \cup A^*)$$

Define the number of routes in $A^* \setminus A$ that contains some u as x_u :

$$x_u = |\{i \in A^* \setminus A, u \in S_i\}|$$

$$u \in T(A^*) \wedge u \notin T(A) \implies x_u \geq 1$$

$$\begin{aligned} \sum_{i \in A^* \setminus A} \Delta(i | A) &= \sum_{i \in A^* \setminus A} [F(A \cup \{i\}) - F(A)] \\ &= \sum_{i \in A^* \setminus A} \left[\sum_{u \in T(A) \cup S_i} a_u - \sum_{u \in T(A)} a_u \right] \\ &= \sum_{i \in A^* \setminus A} \sum_{u \in S_i \setminus T(A)} a_u \\ &= \sum_{u \in T(A^*) \setminus T(A)} x_u a_u \\ F(A \cup A^*) - F(A) &= \sum_{u \in T(A) \cup T(A^*)} a_u - \sum_{u \in T(A)} a_u \\ &= \sum_{u \in T(A^*) \setminus T(A)} a_u \\ F(A \cup A^*) - F(A) &\leq \sum_{i \in A^* \setminus A} \Delta(i | A) \end{aligned}$$

Consider at iteration j , i_j is selected,

$$A_j = \{i_1, i_2, \dots, i_j\}$$

as we pick in order of the highest density,

$$\forall i \in \mathcal{S}, \delta(i \mid A_{j-1}) \leq \delta(i_j \mid A_{j-1})$$

corollarily

$$\forall i \in A^* \setminus A_{j-1}, \delta(i \mid A_{j-1}) \leq \delta(i_j \mid A_{j-1})$$

Using previous result,

$$\begin{aligned} F(A^*) - F(A_{j-1}) &\leq \sum_{i \in A^* \setminus A_{j-1}} \Delta(i \mid A_{j-1}) \\ &= \sum_{i \in A^* \setminus A_{j-1}} \frac{\Delta(i \mid A_{j-1})}{c_i} \cdot c_i \\ &\leq \sum_{i \in A^* \setminus A_{j-1}} \frac{\Delta(i_j \mid A_{j-1})}{c_{i_j}} \cdot c_i \\ &= \frac{\Delta(i_j \mid A_{j-1})}{c_{i_j}} \sum_{i \in A^* \setminus A_{j-1}} c_i \end{aligned}$$

From the budget constraint,

$$\sum_{i \in A^* \setminus A} c_i \leq \sum_{i \in A^*} c_i \leq B$$

we have

$$\begin{aligned} F(A^*) - F(A_{j-1}) &\leq B \cdot \frac{\Delta(i_j \mid A_{j-1})}{c_{i_j}} \\ \Delta(i_j \mid A_{j-1}) &\geq \frac{c_{i_j}}{B} [F(A^*) - F(A_{j-1})] \\ F(A_j) - F(A_{j-1}) &\geq \frac{c_{i_j}}{B} [F(A^*) - F(A_{j-1})] \end{aligned}$$

From the definition of Δ

$$F(A_j) = F(A_{j-1}) + \Delta(i_j \mid A_{j-1})$$

$$\begin{aligned} F(A^*) - F(A_j) &= [F(A^*) - F(A_{j-1})] - [F(A_j) - F(A_{j-1})] \\ &\leq [F(A^*) - F(A_{j-1})] - \frac{c_{i_j}}{B} [F(A^*) - F(A_{j-1})] \\ &= (1 - \frac{c_{i_j}}{B}) [F(A^*) - F(A_{j-1})] \end{aligned}$$

Let the route that causes budget overflow be i_t , Apply previous result up to t , and $\forall c_i, c_i \leq B$

$$\begin{aligned} F(A^*) - F(A_t) &\leq (1 - \frac{c_{i_t}}{B}) [F(A^*) - F(A_{t-1})] \\ &\leq \left(1 - \frac{c_{i_t}}{B}\right) \left(1 - \frac{c_{i_{t-1}}}{B}\right) [F(A^*) - F(A_{t-2})] \\ &\dots \\ &\leq F(A^*) \cdot \prod_{k=1}^t \left(1 - \frac{c_{i_k}}{B}\right) \\ &\leq F(A^*) \cdot \prod_{k=1}^t \left(e^{-\frac{c_{i_k}}{B}}\right) \\ &= F(A^*) \cdot e^{\left(-\sum_{k=1}^t \frac{c_{i_k}}{B}\right)} \end{aligned}$$

Since with the route added in iteration t included, budget constraint is violated,

$$\begin{aligned} \sum_{k=1}^{t-1} c_{i_k} &\leq B < \sum_{k=1}^t c_{i_k}, \quad \frac{-\sum_{k=1}^t c_{i_k}}{B} < -1 \\ F(A^*) - F(A_t) &\leq F(A^*) \cdot e^{\left(\frac{-\sum_{k=1}^t c_{i_k}}{B}\right)} \leq F(A^*) \cdot e^{-1} \\ F(A_t) &\geq \left(1 - \frac{1}{e}\right) \cdot OPT \end{aligned}$$

but $F(A_t)$ is not a feasible solution, as the budget is exceeded. We will discard either the entire greedy prefix to keep the i_t as a singleton collection, or discard i_t to keep A_{t-1}

$$\begin{aligned} ALG &= \max\{F(A_{t-1}), F(\{i_t\})\} \\ &\geq \frac{1}{2} (F(A_{t-1}) + F(\{i_t\})) \\ &\geq \frac{1}{2} F(A_{t-1} \cup \{i_t\}) \\ &= \frac{1}{2} F(A_t) \\ &\geq \frac{1}{2} \left(1 - \frac{1}{e}\right) OPT \end{aligned}$$

Edge case

If no route causes budget overflow, all routes are selected, $ALG = OPT$

2.3 Running time

Outer loop runs at most $|\mathcal{S}|$ times, in each iteration compute Δ takes $|U|$, finding the max among available route takes $|\mathcal{S}|$, overall $O(|\mathcal{S}|^2|U|)$

3 ECS

Question 3: Emergency Communication after a Storm [10 marks]

Following a coastal storm, an emergency team may activate temporary communication relays at candidate sites. Activating site i costs $c_i \geq 0$. A community j can be reached only from sites in a specified set $N(j)$. If no activated relay reaches community j , the team incurs a penalty $p_j \geq 0$ for arranging a slower alternative. Each community can be reached from at most f candidate sites, where $f \geq 1$ is given; that is, $|N(j)| \leq f$ for every community j .

The goal is to minimize

$$\sum_{i \text{ activated}} c_i + \sum_{j \text{ not reached by an activated site}} p_j.$$

- Formulate this problem as an integer linear program and write its natural LP relaxation. [3 marks]
- Using an optimal solution to the LP relaxation, give a deterministic polynomial-time rounding algorithm. [3 marks]
- Prove that your algorithm is an $(f+1)$ -approximation algorithm. [4 marks]

Answer

Goal

Minimize

$$\sum_{i \text{ activated}} c_i + \sum_{j \text{ not reached}} p_j$$

3.1 Integer linear programming formulation

Define relations $S(i)$ as site i is activated; $R(j)$ as community j is not reached.
 Define selection indicator variable x_i as

$$x_i = \begin{cases} 1, & S(i) \\ 0, & \neg S(i) \end{cases} \quad y_j = \begin{cases} 1, & R(j) \\ 0, & \neg R(j) \end{cases}$$

For every j ,

$$\begin{aligned} (\forall i \in N(j) \neg S(i)) \rightarrow R(j) &\equiv \bigwedge_{i \in N(j)} \neg S(i) \rightarrow R(j) \\ &\equiv \sum_{i \in N(j)} (1 - x_i) - y_j \leq |N(j)| - 1 \\ &\equiv |N(j)| - \sum_{i \in N(j)} x_i - y_j \leq |N(j)| - 1 \\ &\equiv \sum_{i \in N(j)} x_i + y_j \geq 1 \end{aligned}$$

Integer constraint: for every x_i, y_j :

$$x_i \in \{0, 1\} \quad y_j \in \{0, 1\}$$

Goal

Minimize

$$\sum_i c_i x_i + \sum_j p_j y_j$$

3.2 LP relaxation

Keep the same objective and covering constraints, and replace the binary constraints by

$$0 \leq x_i \leq 1, \quad 0 \leq y_j \leq 1.$$

Every feasible integer solution is feasible for this relaxation, so $LP \leq OPT$.

3.3 Deterministic rounding

Let the optimal solution of the relaxed LP be x_i^* for i , y_j^* for j , Define a rounding threshold $k \in (0, \frac{1}{f})$:

$$x_i = \begin{cases} 1, & x_i^* \geq k \\ 0, & \text{otherwise} \end{cases}$$

This rounding is clearly polynomial, linear to the size of site set.

3.4 Approximation factor proof

Using this deterministic rounding, in our result $\forall i, S(i) \rightarrow$

$$\begin{aligned} x_i^* &\geq k \\ \frac{x_i^*}{k} &\geq 1 \\ \frac{1}{k} c_i x_i^* &\geq c_i x_i \quad (x_i \leq 1) \\ \Rightarrow \frac{1}{k} \sum_i c_i x_i^* &\geq \sum_i c_i x_i \end{aligned}$$

For each j that is unreachable, we need to have

$$\begin{aligned}\forall i \in N(j)(\neg S(i)) &\implies \forall i \in N(j)(x_i^* < k) \\ &\implies \sum_{i \in N(j)} x_i^* \leq k|N(j)| \leq fk\end{aligned}$$

From constraint,

$$\begin{aligned}y_j^* &\geq 1 - \sum_{i \in N(j)} x_i^* \\ y_j^* &\geq 1 - fk \\ \frac{1}{1-fk} \cdot y_j^* &\geq 1 \\ \frac{1}{1-fk} \cdot p_j y_j^* &\geq p_j y_j \quad (y_j \leq 1) \\ \frac{1}{1-fk} \sum_j p_j y_j^* &\geq \sum_j p_j y_j\end{aligned}$$

Combining 2 parts, we have

$$\begin{aligned}LP &= \sum_i c_i x_i^* + \sum_j p_j y_j^* \\ ALG &= \sum_i c_i x_i + \sum_j p_j y_j \leq \frac{1}{k} \sum_i c_i x_i^* + \frac{1}{1-fk} \sum_j p_j y_j^* \leq \max\left\{\frac{1}{k}, \frac{1}{1-fk}\right\} LP\end{aligned}$$

Find k to minimize $\max\left\{\frac{1}{k}, \frac{1}{1-fk}\right\}$

Notice that $g_1(k) = \frac{1}{k}$ is strictly decreasing, $g_2(k) = \frac{1}{1-fk}$ is strictly increasing over $k \in (0, \frac{1}{f})$ ($k = 0$ case is trivially impossible), for $f \geq 1$, $\max\left\{\frac{1}{k}, \frac{1}{1-fk}\right\}$ occurs at the intersection when

$$\begin{aligned}\frac{1}{k} &= \frac{1}{1-fk} \\ k &= \frac{1}{f+1}\end{aligned}$$

Given

$$\begin{aligned}ALG &\leq \max\left\{\frac{1}{k}, \frac{1}{1-fk}\right\} LP \\ ALG &\leq (f+1) \cdot LP \leq (f+1) \cdot OPT\end{aligned}$$

3.5 Running time

Let n denote the total input size, suppose the relaxed LP is solved in $O(n^a)$ time, where $a \geq 1$ is a constant, rounding all site variables and checking which communities are reached takes

$$O\left(|I| + |J| + \sum_{j \in J} |N(j)|\right) = O(n)$$

additional time. Therefore, the overall running time is $O(n^a)$, determined by the LP solver